

Study Protocol

Title: Water Source and Diarrhoeal Disease: A Prospective Cohort Study in a Simulated District Population

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 - **Principal Investigator:** Arsénio Nhacolo
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1. Background and Rationale

Diarrhoeal diseases remain a leading cause of morbidity and mortality worldwide, disproportionately affecting young children in low- and middle-income countries. A primary and modifiable risk factor for these diseases is the lack of access to safe drinking water and adequate sanitation infrastructure. Despite widespread recognition of this association, quantifying the magnitude of risk attributable to specific water sources in local contexts remains a public health priority.

This study was designed to provide a realistic epidemiological scenario for the AMMnet Lusophone Hackathon. By simulating a prospective cohort study in a defined district population, participants gain hands-on experience in applying core epidemiological measures (incidence proportion, incidence rate, and measures of association) to a research question of direct public health relevance.

2. Study Objectives

2.1 Primary Objective

To estimate and compare the incidence of diarrhoea among individuals using well water versus tap water as their primary household water source.

2.2 Secondary Objectives

1. To describe the baseline sociodemographic and household characteristics of the study cohort, stratified by water source.
2. To estimate the overall incidence proportion (cumulative incidence) of diarrhea, defined as the risk of experiencing at least one episode during the follow-up period.
3. To estimate the overall incidence rate of diarrhoea, accounting for recurrent episodes and total person-time at risk.
4. To evaluate the crude (unadjusted) association between primary water source and both the risk and rate of diarrhoea using the Risk Ratio (RR) and the Incidence Rate Ratio (IRR).
5. To explore potential confounding variables and discuss sources of bias.

3. Study Design

This is a **prospective cohort study** simulating 12 months of follow-up in a defined district population. Participants will be enrolled and classified at baseline according to their primary household water source (the exposure). All enrolled participants will then be followed prospectively over time to record the occurrence and frequency of diarrhoeal episodes (the outcome).

The prospective cohort design was chosen because it allows:

- Direct estimation of disease incidence in exposed and unexposed groups.
- Capture of multiple (recurrent) outcome events per participant.
- Calculation of person-time at risk, enabling incidence rate estimation.
- Assessment of temporality, the exposure precedes the outcome.

4. Study Population

4.1 Source Population

Residents of a specific administrative district, selected to represent a population with heterogeneous access to water and sanitation infrastructure.

4.2 Eligibility Criteria

Inclusion criteria:

- Permanent resident of the study district at the time of enrolment.
- Age ≥ 1 year at enrolment.
- Using either tap water or well water as the primary household drinking water source.
- Willing and able to provide informed consent (or assent with parental consent for minors).

Exclusion criteria:

- Temporary residents or visitors to the district.
- Use of a water source other than tap water or well water as the primary source.
- Inability to be followed up (e.g., planned relocation within the study period).

4.3 Enrolment Period

Enrolment is planned to take place between **1 January 2024** and **31 March 2024**. The study targets the enrolment of approximately **1,200 participants** across this period.

5. Exposure and Outcome Definitions

5.1 Primary Exposure: Water Source

The primary exposure is the **type of household drinking water source** at the time of enrolment, classified as a binary variable:

- **Tap water (coded 0):** Piped water supplied to the household, presumed to be treated.
- **Well water (coded 1):** Water drawn from a groundwater well (protected or unprotected), presumed to be untreated or minimally treated.

Exposure status will be determined at enrolment and treated as **fixed** throughout the follow-up period (i.e., no exposure switching is planned).

5.2 Primary Outcome: Diarrhoeal Disease

The primary outcome is the occurrence of **diarrhoeal episodes** during the 12-month follow-up period, captured in two complementary ways:

- **Binary outcome (for incidence proportion):** Whether the participant experienced at least one diarrhoeal episode during follow-up (yes/no).
- **Count outcome (for incidence rate):** The total number of diarrhoeal episodes experienced during follow-up.

For the purposes of this study, a diarrhoeal episode is assumed to last an **average of 3 days**. This assumption will be used in the calculation of net person-time at risk (see Section 9).

6. Data Collection

At **enrolment**, the following baseline variables will be collected for each participant via structured household interview:

Variable	Description
Participant ID	Unique sequential numeric identifier
Enrolment date	Date of enrolment into the study
Age	Age in whole years at the date of enrolment
Sex	Biological sex (male/female)
Primary water source	Tap water or well water
Sanitation access	Whether the household has access to improved sanitation facilities
Household size	Total number of people living in the household
Socioeconomic status	Household socioeconomic classification (low, medium, high)
Education of household head	Highest education level attained by the household head

During follow-up, the following outcome and follow-up data will be recorded:

Variable	Description
Number of diarrhoeal episodes	Total count of episodes during the follow-up period
Time to first episode	Days from enrolment to the first diarrhoeal episode (missing if no episode occurred)
Total follow-up time	Number of days from enrolment to the end of follow-up or to censoring
Lost to follow-up	Whether the participant was lost to follow-up before the end of the study period

7. Sample Size

The study targets a sample size of approximately **1,200 participants**. This sample size is designed to provide adequate statistical power to detect a meaningful difference in diarrhoea incidence between the two exposure groups (tap water vs. well water), while remaining feasible for a district-level study.

8. Follow-up Procedures

8.1 Follow-up Duration

The maximum follow-up period will be **365 days** (approximately 12 months) from the date of enrolment. Participants enrolled between January and March 2024 will be followed until their corresponding date in 2025, or until censoring (whichever comes first).

8.2 Censoring

A participant's follow-up time will be **censored** (i.e., their observation ends before the maximum 365 days) if they are **lost to follow-up** (e.g., due to relocation or drop-out). Their follow-up time will reflect the number of days of observation before they were lost.

8.3 Recurrent Events

Because participants may experience **more than one diarrhoeal episode** during follow-up, each event will be recorded to capture the total count of episodes per participant over the entire follow-up period, as well as the time to the initial event.

9. Statistical Analysis Plan

9.1 Descriptive Analysis

Baseline characteristics of the cohort will be described and compared between the two exposure groups (tap water vs. well water) overall and in a stratified Table 1 generated via `gtsummary::tbl_summary()`:

- **Continuous variables** (age, household size): mean (SD) or median (IQR) based on data distribution, compared using the independent-samples t-test or Wilcoxon rank-sum test.
- **Categorical variables** (sex, sanitation, socioeconomic status, education of household head): frequency (n) and percentage (%), compared using the chi-squared test or Fisher's exact test.
- Missing values will be explicitly quantified and reported for all variables.

9.2 Incidence Proportion (Risk)

A binary outcome variable representing diarrhoea status will be derived, indicating whether a participant experienced at least one diarrhoeal episode during follow-up.

- **Overall incidence proportion** = number of participants with at least one diarrhoeal episode / total number of participants.
- **95% confidence interval** will be calculated using the exact binomial (Clopper-Pearson) method.

- The risk in the exposed group (well water) will be compared to the unexposed group (tap water) using the Risk Ratio (RR) with 95% Wald CI.

9.3 Incidence Rate

Person-time at risk will be calculated for each participant to account for recurrent episodes and censoring/loss to follow-up:

1. **Net person-time (days)** = total follow-up time – (number of diarrhoeal episodes × 3)
2. **Person-time (years)** = net person-time (days) / 365.25
3. **Incidence rate** = total number of episodes / total person-time (years), expressed per 1,000 person-years.
4. **95% confidence interval** will be derived assuming a Poisson distribution.
5. The rate in the exposed group will be compared to the unexposed group using the Incidence Rate Ratio (IRR) with 95% Poisson CI.

9.4 Multivariable Modeling & Confounder Adjustment

To control for potential confounding variables, multivariable regression models will be executed. Covariates adjusted for will include: age, sex, access to improved sanitation, household socioeconomic status, and education of the household head. Stepwise model selection will be avoided.

- **Diarrhoea Risk (Binary):** Multivariable logistic regression will estimate adjusted Odds Ratios (aOR) and 95% profile-likelihood CIs.
- **Recurrent Episode Rates (Count):** Negative Binomial regression with an offset of the logarithm of person-time (in years) will estimate adjusted Incidence Rate Ratios (aIRR). Robust standard errors will be computed to account for within-person correlation of recurrent events.

9.5 Survival (Time-to-Event) Analysis

To analyze the time from study enrolment to the first diarrhoea episode:

- **Kaplan-Meier Estimator:** Survival curves stratified by water source will estimate diarrhoea-free probability over follow-up days. Stratified comparison will use the log-rank test.
- **Cox Proportional Hazards Model:** Used to estimate crude and adjusted Hazard Ratios (cHR, aHR) and 95% CIs. The proportional hazards assumption will be tested using Schoenfeld residuals via `survival::cox.zph()`.
- **Collinearity:** Variance Inflation Factors (VIF) will be calculated to identify multicollinearity.

9.6 Software

All analyses will be performed in **R** (version ≥ 4.0). Key R packages include: `here`, `readr`, `dplyr`, `gtsummary`, `broom`, `survival`, and `survminer`.

10. Ethical Considerations

This study is based on **simulated data** and does not involve real human participants. Therefore, formal ethical review and informed consent procedures are not applicable in the context of the AMMnet Lusophone Hackathon.

In a real-world application of this study design, the following ethical principles would apply:

- Written informed consent (or parental/guardian consent for minors) would be required from all participants prior to enrolment.
- All personal identifiers would be replaced with pseudonymous codes in the analysis dataset.
- Participants would retain the right to withdraw from the study at any time without penalty.
- Data would be stored securely in compliance with applicable data protection regulations.
- Results would be disseminated to the study community and relevant health authorities.